ECE 501: Digital Image Processing

Course Project – Week 6 Progress Report

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

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Project 17, Group 7

Project Title

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Team Details

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1 Work Done This Week

1.1 Fabric-ID Parsing Fixed (Correction to Earlier Reports)

We inspected the real filenames and corrected the parser. AFID names have the form `<image number>_<defect-type code>_<fabric code>` (e.g. `0092_030_01`). Over the 103 defective images the three tokens take 103, 12 and 7 distinct values, which match the 12 defect types and 7 fabric types of AFID. Our earlier parser took the first token (the image number) as the "fabric ID" and the last token (the fabric code) as the "defect code". This has three consequences for earlier reports:

- The counts of 20 defect-free vs. 106 defective fabric IDs (17 shared), and the explanation that the pooled model was needed because fabric IDs had no defect-free counterpart, were artifacts of the wrong token. With the correct parse, all 7 fabric types have both defect-free and defective images (Table 2).
- The "defect code" plot in the mid-semester report (Fig. 1a) and the "AUC by defect type" plot in the Week 5 report (Fig. 3) were actually split by *fabric* type. The same four groups with 405, 40, 216 and 16 defective patches reappear below as fabrics 01–04 (Table 8).
- The true defect-type breakdown is given in Table 8.

1.2 Label Analysis: Strict vs. Relaxed Defect Labels

We recomputed the exact overlap between every patch and its ground-truth mask. Under the strict rule used so far (patch is defective if at least 10% of its pixels are masked), there are 677 defective patches (0.53%), but they come from only **15 of the 103 defective images**. Under a relaxed rule (any overlap) there are 2,087 defective patches (1.64%) from 102 of 103 images. The strict positives are also concentrated: a single image of defect type 036 provides 379 of the 677 (56%), and defect types 010, 006 and 002 provide 216, 54 and 26; every other defect type, including the most frequent type 019 (38 images), has fewer than five strict-positive patches (Table 3). The earlier descriptor ranking (Week 5) was therefore driven by the larger defects of a few images. We now report both label definitions and treat the relaxed one as the more representative benchmark.

1.3 Per-Fabric Models and Thresholds

Using the corrected fabric IDs we ran image-level cross-validation (5 folds, 3 seeds, no patches of one image shared between training and test) with a pooled model and a per-fabric model, on the selected Gabor + FFT features (24 dimensions) and on all 68 features. No fabric needed the pooled fallback. We also compared a global score threshold with a per-fabric threshold, and tested morphological post-processing of the predicted patch grid. Thresholds are set from held-out scores of defect-free patches at a target false-positive rate (FPR) of 1%, 2% or 5%.

1.4 Defect-Level Evaluation

Pixel IoU is almost uninformative for thin defects (oracle IoU is 0.03, Week 5). We therefore report a defect-level measure: *defect recall* is the share of defective images in which at least one flagged patch overlaps the ground-truth defect, and false alarms are the flagged patches with no ground-truth overlap, per image, together with the share of defect-free images with at least one flagged patch. Pixel IoU is still listed for reference.

1.5 Channel Check

We checked the format of all 143 images: 129 are single-channel 8-bit and 14 have four channels. All images are loaded as grayscale, so the pipeline runs, but we have not yet checked which images have four channels or whether the extra channel is constant.

1.6 Preprocessing Ablation

Preprocessing (CLAHE with clip limit 2.0 and 8×8 tiles, followed by a 3×3 Gaussian blur) had been chosen in Week 2 and never tested. We compared seven variants: raw grayscale, blur only, median filter (3×3), CLAHE only, CLAHE + blur (the current choice), illumination flattening (division by a Gaussian-smoothed copy with $\sigma = 24$ px, rescaled to the image mean), and flattening + blur. Each variant recomputes the 24 Gabor + FFT features on the same patch grid and is evaluated with the same per-fabric model, labels and image-level folds, so only the preprocessing differs. As a check, the recomputed features for the current preprocessing reproduce the cached ones exactly (maximum difference 0). We also measured illumination variation as the brightness variation between images of the same fabric and as the brightness gradient along the strip (coefficient of variation of the column means, which includes the image borders). Because fold-level comparisons over three seeds are not independent, we added a bootstrap over images for the two flattening variants.

1.7 Reproducible Defect-Free Sample (Which Result Uses Which Data)

The cached features used only 40 of the 141 defect-free images, sampled from an unsorted file list, so the sample was not the same in every Kaggle session. We noticed this when a recomputation on a rebuilt cache gave defect recall 0.167 and 0.353 at 1% and 5% FPR (17 and 36 of 102 defective images) instead of 0.175 and 0.379 (18 and 39 of 103), and the number of defect-free images per fabric (00–06) changed from 7/5/6/4/4/4/10 to 5/5/4/6/4/7/9. We fixed the build (sorted file list) and now use *all* 141 defect-free images (20 or 21 per fabric). Because results were produced before and after the fix, Table 1 states which sample each result uses; re-running everything on the final sample is the first item of Work To Be Done, and individual numbers in the earlier sections may shift slightly when it is done.

Table 1: Defect-free sample used for each group of results (all samples use the same 103 mask-matched defective images)

Sample	Defect-free images (fabrics 00–06)	Used for
A	40 (7/5/6/4/4/4/10)	Sections 1.1–1.6 and Tables 2–10 (parser fix, labels, per-fabric models, defect-level, post-processing, pre-processing)
B	40 (5/5/4/6/4/7/9)	Qualitative examples and localization test (Section 2.6)
C	141 (20/21/20/20/20/20/20)	Image-level detection (Section 2.7)

1.8 Qualitative Examples and Localization Test (Sample B)

To see what the pipeline actually finds, we produced example outputs (anomaly score map and flagged patches over the image, with the ground truth outlined) for images chosen by fixed rules, not by hand: the cleanest detections (found at 1% FPR with the fewest false alarms), borderline cases (found only at 5% FPR) and the largest missed defects. Of 102 defective images (one mask overlaps no patch and is excluded) the defect is found at 1% FPR in 17 (0.167), in 36 at 5% FPR (0.353), and not found even at 5% in 66. Our "found" criterion is lenient: any flagged patch that touches the defect counts, even when many other patches are flagged. Observations from the examples:

- **Real localization on thin defects.** In two detected images (defects covering 0.04% and 0.07% of the image, thin curved lines) the anomaly score peaks at the defect and the flagged patch touches it, with 2 and 6 false-alarm patches.
- **A large structural defect** (13.7% of the image, a band along the top of the strip) is flagged over its full width.
- **Detections that are partly luck.** In a borderline image the flagged region covers a bright seam and a dark area (60 false-alarm patches) and reaches the tiny defect only at its edge; in another, 281 patches (about a third of the image) are flagged, so touching the defect is not meaningful.
- **Misses.** Long, faint, one- or two-pixel lines are not visible in the score map. In one missed image the score does peak at the defect but stays below the 5% threshold.
- **False alarms** come from fabric edges, seams, blank white margins and, in some images, a higher score level over the whole strip.

Because "found" is lenient, we added a stricter test that does not depend on any threshold: patches are ranked within each image and we check whether the top-1, top-3 or top-10 patches overlap the defect, compared with the rate expected from randomly chosen patches.

1.9 Image-Level Detection (Sample C)

We also asked the whole-strip question: is a strip defective or not? Each image receives one score computed from its out-of-fold patch scores (Gabor + FFT, CLAHE + blur, per-fabric model), and we measure how well the score separates the 103 defective images from the 141 defect-free images (AUC with an image bootstrap, recall at fixed false-positive rates measured on the defect-free images, and precision at assumed prevalences, since 42% defective images in this set is not a realistic base rate). Four ways of combining patch scores (maximum, mean of the top five, maximum minus the image median, maximum local contrast) and eight whole-image statistics were compared. Because defective images could differ from defect-free ones for reasons unrelated to defects, we added controls: the score computed only on patches far from the defect, image brightness, contrast and blank fraction as alternative predictors, the AUC within each fabric, and a cross-validated logistic regression testing whether the score adds information beyond fabric and brightness.

2 Results

Table 2: Images per fabric type after the parser fix (the 40 defect-free images used and all 103 mask-matched defective images)

Fabric code	00	01	02	03	04	05	06
Defect-free	7	5	6	4	4	4	10
Defective	7	12	38	30	9	1	6

Table 3: Defective images per defect-type code, and strict-label defective patches (only counts of at least five are shown)

Defect code	Images	Strict defective patches
019	38	< 5
030	13	< 5
010	10	216
022	9	< 5
002	9	26
006	8	54
025	5	< 5
023	5	< 5
016	3	< 5
029, 036, 027	1 each	036: 379; others < 5

2.1 Patch-Level Results

Table 4: Patch-level average precision (AP) and ROC-AUC, image-level cross-validation (13 folds with positives out of 15 for strict labels). Chance AP: 0.0053 strict, 0.0164 relaxed.

Features	Model	AP strict	AUC strict	AP relaxed	AUC relaxed
Gabor + FFT (24)	pooled	0.119	0.895	0.104	0.754
Gabor + FFT (24)	per-fabric	**0.180**	0.919	**0.141**	**0.796**
All four (68)	pooled	0.098	0.883	0.086	0.748
All four (68)	per-fabric	0.113	**0.934**	0.099	**0.796**

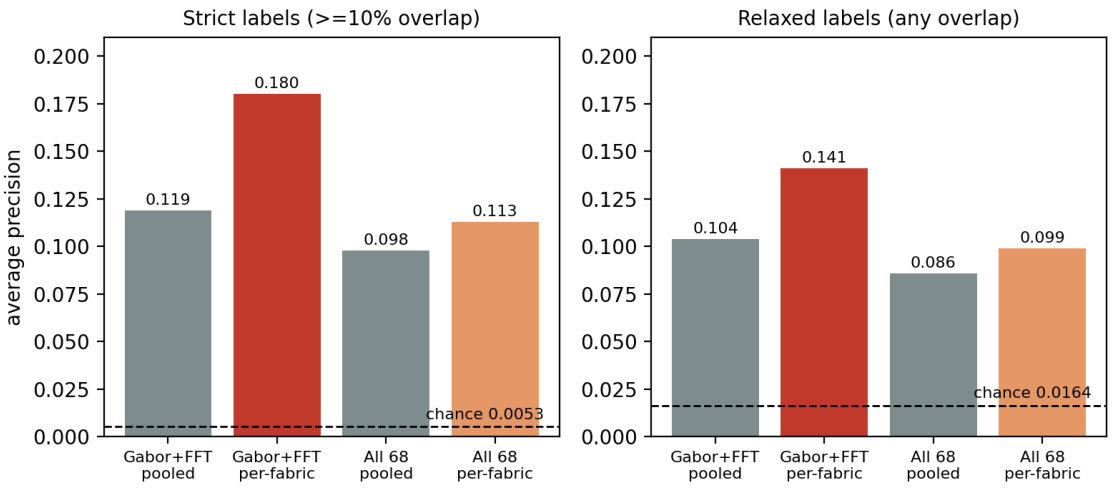


Figure 1: Patch-level average precision for pooled and per-fabric models under strict and relaxed labels.

Paired comparisons over folds (strict labels): per-fabric beats pooled for Gabor + FFT by ΔAP = +0.061 (wins in 77% of 13 folds) and for all features by +0.015 (69%). Gabor + FFT beats all four descriptors by +0.021 with the pooled model (77%) and by +0.067 with per-fabric models (69%), but the all-feature model has the higher strict AUC (0.934 vs 0.919) and the two have equal relaxed AUC (0.796).

2.2 Defect-Level Results

Table 5: Defect-level results, per-fabric threshold, no post-processing. Recall = defective images where a flagged patch touches the defect; "flagged" = share of defect-free images with at least one false alarm.

Features / model	FPR	Defect recall	False alarms/img	Flagged normal	Pixel IoU
Gabor + FFT, pooled	1%	0.155	8.8	0.850	0.014
	2%	0.214	17.5	0.975	0.016
	5%	0.252	43.7	1.000	0.013
Gabor + FFT, per-fabric	1%	0.175	8.8	0.650	0.018
	2%	0.204	17.5	0.675	0.015
	5%	**0.379**	43.7	0.950	0.014
All four, pooled	1%	0.126	8.8	0.850	0.008
	2%	0.194	17.5	1.000	0.012
	5%	0.301	43.7	1.000	0.014
All four, per-fabric	1%	0.175	8.8	0.625	0.017
	2%	0.223	17.5	0.650	0.015
	5%	0.340	43.7	0.675	0.014

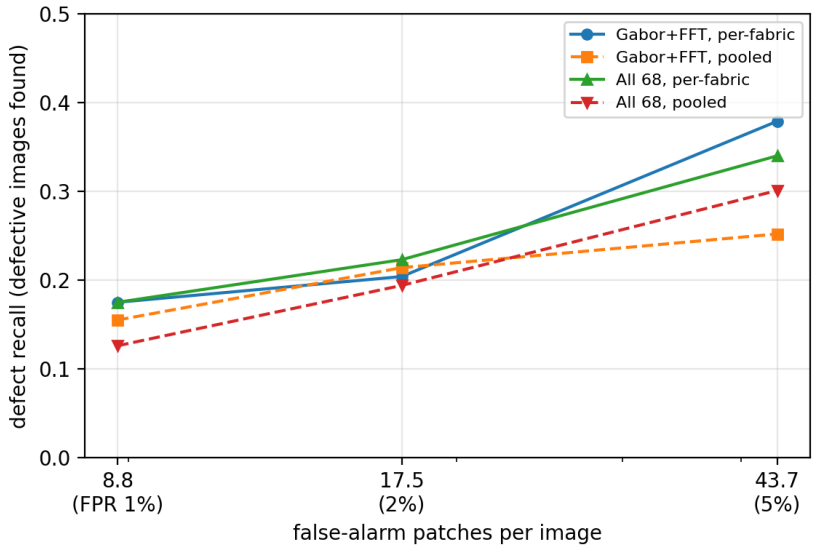


Figure 2: Defect recall against false alarms per image for the four model variants (per-fabric thresholds, no post-processing).

Table 6: Global vs. per-fabric threshold (per-fabric model, Gabor + FFT, no post-processing)

FPR	Threshold	Defect recall	False alarms/img	Flagged normal	Pixel IoU
1%	global	0.097	8.7	0.475	0.018
1%	per-fabric	0.175	8.8	0.650	0.018
2%	global	0.165	17.5	0.775	0.018
2%	per-fabric	0.204	17.5	0.675	0.015
5%	global	0.311	43.7	1.000	0.015
5%	per-fabric	0.379	43.7	0.950	0.014

2.3 Morphological Post-Processing

Table 7: Post-processing of the flagged-patch grid (per-fabric model, Gabor + FFT, per-fabric threshold). open2: opening with a 2×2 element; close3: closing with a 3×3 element.

FPR	Post-processing	Defect recall	False alarms/img	Flagged normal	Pixel IoU
1%	none	0.175	8.8	0.650	0.018
1%	open2	0.126	4.3	0.100	0.014
1%	close3	0.155	7.2	0.650	0.011
1%	open2 + close3	0.126	3.4	0.100	0.011
5%	none	0.379	43.7	0.950	0.014
5%	open2	0.214	29.8	0.725	0.013
5%	close3	0.291	37.7	0.925	0.009
5%	open2 + close3	0.204	23.1	0.725	0.010

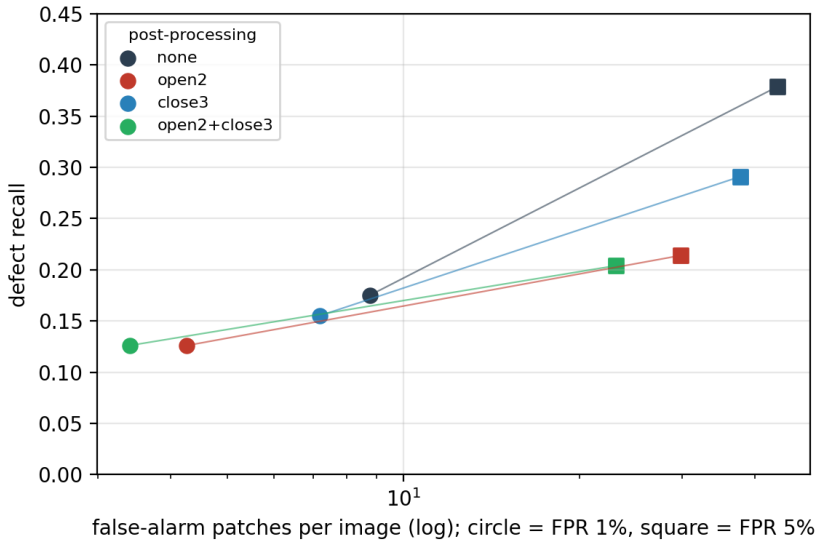


Figure 3: Effect of post-processing on recall and false alarms (Gabor + FFT, per-fabric model).

2.4 AUC by Fabric Type and by True Defect Type

Table 8: Patch-level AUC (per-fabric model, Gabor + FFT, strict labels); groups with fewer than five strict-positive patches are omitted.

Fabric	Patches	AUC	Defect type	Patches	AUC
01	405	0.978	002	26	0.839
02	40	0.961	006	54	0.975
03	216	0.939	010	216	0.921
04	16	0.977	036	379	0.986

2.5 Preprocessing Ablation

Table 9: Preprocessing variants (per-fabric model, Gabor + FFT, image-level cross-validation). AP chance level: 0.0164 relaxed, 0.0053 strict. Recall and false alarms use per-fabric thresholds; at 1% FPR every variant gives 8.8 false-alarm patches per image. Illumination columns: coefficient of variation of image brightness between images of a fabric (between) and along the strip (gradient).

Variant	AP rel.	AUC rel.	AP strict	AUC strict	Recall 1%	Recall 5%	Between	Gradient
Raw	0.095	0.784	0.100	0.890	0.117	0.330	0.024	0.316
Blur	0.102	0.780	0.119	0.876	0.136	0.350	0.024	0.316
Median	0.090	0.783	0.093	0.876	0.117	0.340	0.024	0.319
CLAHE	0.124	0.800	0.152	0.934	0.175	0.350	0.020	0.266
CLAHE + blur (current)	0.141	0.796	0.180	0.919	0.175	**0.379**	0.020	0.265
Flatten	0.154	0.801	0.207	0.939	0.204	0.340	0.024	0.039
Flatten + blur	**0.163**	0.799	**0.218**	0.932	0.204	0.350	0.024	**0.035**

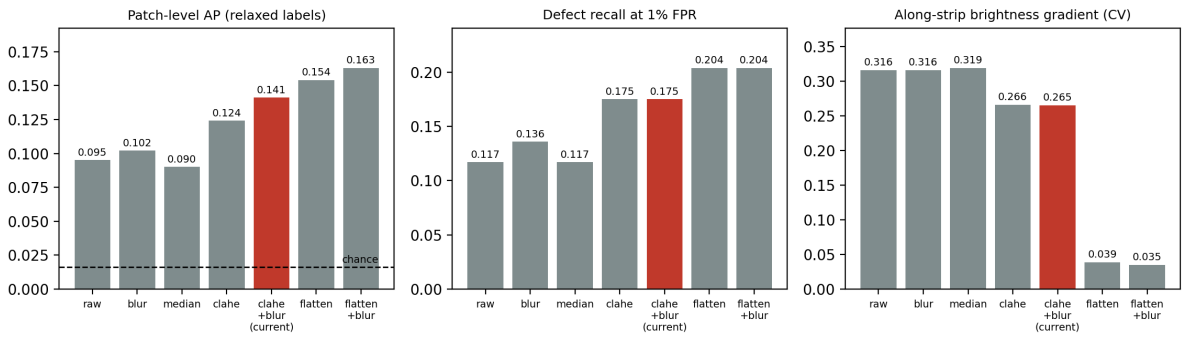


Figure 4: Effect of preprocessing on patch-level AP, defect recall at 1% FPR and along-strip brightness gradient (current choice in red).

Table 10: Comparison with the current preprocessing (CLAHE + blur) in relaxed-label AP. Folds: 15 (3 seeds × 5 folds, not independent). Bootstrap: 300 resamples of images, seed-0 out-of-fold scores.

Variant	ΔAP vs current	Better in % of folds	Bootstrap 95% CI of ΔAP
Raw	−0.046	33	–
Blur	−0.039	27	–
Median	−0.051	33	–
CLAHE	−0.018	53	–
Flatten	+0.013	67	[−0.022, +0.090] (median +0.005)
Flatten + blur	+0.022	60	[−0.015, +0.098] (median +0.041)

2.6 Qualitative Examples and Localization Test (Sample B)

Table 11: Detection by defect-type code (Sample B; 102 defective images). Recall = defective images where a flagged patch touches the defect.

Defect code	Images	Recall at 1% FPR	Recall at 5% FPR	Mean defect area (% of image)
019	38	0.000	0.132	0.003
030	13	0.000	0.154	0.015
002	9	0.000	0.111	0.100
010	9	0.333	0.778	2.300
022	9	0.111	0.667	0.029
006	8	1.000	1.000	0.191
025	5	0.000	0.000	0.002
023	5	0.200	0.600	0.087
016	3	1.000	1.000	0.078
027, 029	1 each	0.000	0.000	0.027, 0.029
036	1	1.000	1.000	22.268

Table 12: Localization test: share of defective images in which one of the k highest-scoring patches of the image overlaps the defect (102 images), against the rate expected from random patches

	Top-1	Top-3	Top-10
Pipeline	0.108	0.137	0.216
Random patches	0.023	0.058	0.141

Defect size	Images	Top-1	Top-10	Random top-10
$< 0.01\%$ of image	51	0.000	0.020	0.044
0.01–0.1%	27	0.111	0.222	0.109
$> 0.1\%$	24	0.333	0.625	0.383

2.7 Image-Level Detection (Sample C)

Table 13: Image-level AUC of four ways to combine patch scores (103 defective vs 141 defect-free images; bootstrap 95% CI) and recall at fixed FPR on the defect-free images

Image score	AUC [95% CI]	Recall 5% FPR	Recall 10% FPR	Recall 20% FPR
Maximum patch score	0.497 [0.43, 0.57]	0.087	0.136	0.214
Mean of top five	0.502 [0.42, 0.58]	0.117	0.175	0.214
Maximum − image median	0.396 [0.33, 0.46]	0.087	0.107	0.136
Maximum local contrast	0.394 [0.32, 0.45]	0.058	0.087	0.136

Table 14: Whole-image statistics as image-level scores (AUC with bootstrap 95% CI). "Far from defect" excludes patches within about 130 px of the ground-truth defect (analysis only, not a usable detector). Mean brightness: 127.4 (defective) vs 125.7 (defect-free).

Statistic	AUC [95% CI]
Share of patches above the per-fabric 95th-percentile threshold	**0.766** [0.70, 0.83]
Same share, far from the defect	0.707 [0.63, 0.77]
90th percentile of patch score	0.691 [0.62, 0.76]
Mean patch score	0.639 [0.57, 0.71]
Image brightness	0.626 [0.56, 0.69]
Mean patch score, far from the defect	0.589 [0.52, 0.66]
Image contrast	0.481 [0.41, 0.56]
Blank (white) fraction	0.459 [0.39, 0.53]

Table 15: Operating points of the best image-level scores. Precision is computed for an assumed share of defective strips (10% and 50%) from recall and false-positive rate.

Score	AUC	Recall 5% FPR	Recall 10% FPR	Recall 20% FPR	Prec., 10% defective	Prec., 50% defective
Share > threshold	0.766	0.379	0.515	0.631	0.365	0.838
90th percentile	0.691	0.359	0.456	0.515	0.338	0.821
Mean score	0.639	0.184	0.369	0.437	0.292	0.788

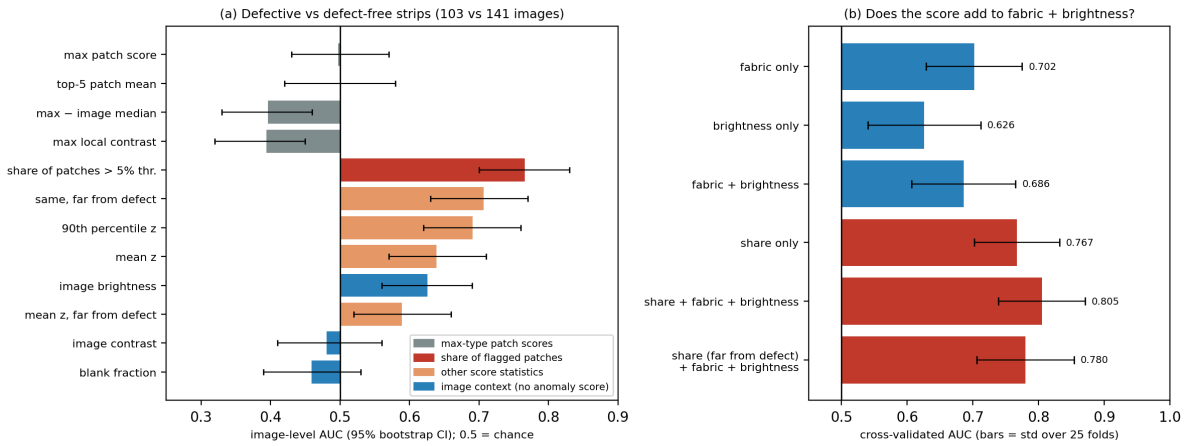


Figure 5: (a) Image-level AUC of patch-score combinations and whole-image statistics; the maximum-type scores are at or below chance, the share of flagged patches reaches 0.77. (b) Cross-validated AUC of a logistic regression with and without the share score.

Table 16: AUC within each fabric (fabrics with at least 3 images of each class; fabric 05 has only one defective image)

Fabric	Defective	Defect-free	AUC, share score	AUC, brightness
00	7	20	0.639	0.386
01	12	21	0.895	0.738
02	38	20	0.807	0.608
03	30	20	0.837	0.495
04	9	20	0.406	0.311
06	6	20	0.633	0.633

Table 17: Cross-validated AUC of a logistic regression (5-fold, 5 repeats; std over the 25 folds)

Features	CV AUC	std
Fabric only	0.702	0.073
Brightness only	0.626	0.086
Fabric + brightness	0.686	0.079
Share score only	0.767	0.065
Share score + fabric + brightness	**0.805**	0.066
Share score (far from defect) + fabric + brightness	0.780	0.074

2.8 Interpretation

- **The fabric-ID hypothesis is supported.** Per-fabric models improve patch-level AP over the pooled model (Gabor + FFT: 0.180 vs 0.119 strict, 0.141 vs 0.104 relaxed), and per-fabric thresholds give higher defect recall than a global threshold at the same number of false alarms (e.g. 0.175 vs 0.097 at 1% FPR).

- **Gabor + FFT is at least as good as all 68 features.** It has the higher AP in every setting, with a 24-dimensional feature vector, but the all-feature model has equal or higher AUC, so we describe the reduced set as comparable and cheaper rather than clearly better.

- **Absolute performance on thin defects is still poor.** Under relaxed labels the AUC is about 0.80, against 0.92–0.93 under strict labels, so the strong strict-label numbers mainly reflect larger defects. At the defect level the best setting finds 37.9% of defective images at

5% FPR, with 43.7 false-alarm patches per image and 95% of defect-free images flagged; at 1% FPR recall is 17.5%. Pixel IoU stays at about 0.014–0.018.

- **Opening reduces false alarms at a cost in recall; closing does not help.** At 1% FPR, opening with a 2×2 element cuts false alarms from 8.8 to 4.3 per image and the share of flagged defect-free images from 65% to 10%, while recall falls from 0.175 to 0.126. At 5% FPR the same operation reduces recall from 0.379 to 0.214. Closing lowers IoU and removes few false alarms.

- **CLAHE + blur is clearly better than no preprocessing.** Raw, blur-only and median preprocessing all have lower AP (0.090–0.102 vs 0.141 relaxed) and are better than the current choice in only 27–33% of folds. Flattening is the best-scoring alternative (AP 0.154 and 0.163, strict AP 0.207 and 0.218) and removes most of the along-strip brightness gradient (0.265 $\rightarrow$ 0.035–0.039), but the image bootstrap cannot separate it from CLAHE + blur (both intervals include zero), AUC is unchanged (0.796–0.801), and at 5% FPR the current choice has the higher recall (0.379 vs 0.340–0.350). We therefore keep CLAHE + blur and treat flattening as a promising, unconfirmed candidate. Illumination varies mainly along the strip (gradient 0.32 raw) rather than between images of a fabric (0.024), although the gradient measure also includes the image borders.

- **The pipeline localizes some defects but does not detect them exactly (Sample B).** Defects are found in 17 of 102 images at 1% FPR and 36 at 5%, and the lenient "found" criterion overstates this. Detection follows defect size: types with large mean area (006, 010, 036) are found most often, while the most frequent type 019 (38 images, mean area 0.003% of the image, about 30 pixels) is found in 0% and 13% of images (Table 11). In the stricter test the highest-scoring patch lies on the defect in 10.8% of images (random: 2.3%) and a top-10 patch in 21.6% (random: 14.1%). For defects under 0.01% of the image, the top-10 rate (2.0%) is below random (4.4%): the highest scores there are on seams, edges and margins, not on the defect.

- **Image-level detection fails with maximum-type scores but works moderately with the share of flagged patches (Sample C).** Maximum, top-five mean and local-contrast scores are at or below chance (AUC 0.39–0.50, the last two significantly below). The share of patches above the per-fabric 95th-percentile threshold gives AUC 0.77 [0.70, 0.83], with recall 0.38, 0.52 and 0.63 at 5%, 10% and 20% FPR (Table 15). At 10% defective strips and 10% FPR, about 36% of flagged strips would be defective, and 84% at 50% defective. The 42% base rate in this set makes high "precision on this set" values meaningless, so we do not report them.

- **The image-level signal is not just fabric or brightness, but it is not uniform.** Fabric alone gives AUC 0.70 (defective images are concentrated in some fabrics, e.g. fabric 05 has one defective and 20 defect-free images) and brightness 0.63, but adding the share score raises the cross-validated AUC from 0.686 to 0.805 (Table 17; fold-to-fold std about 0.07). Within fabrics the score separates well for fabrics 01–03, which hold most defective images (AUC 0.81–0.90), and poorly for 00, 04 and 06 (0.64, 0.41, 0.63; Table 16).

- **Defective strips score higher even far from the defect.** Excluding patches near the defect the share score keeps AUC 0.71, and clean patches of defective images are flagged about four times as often as patches of defect-free images (1.8% vs 0.4% at the 1% threshold). This could be a real texture change over the strip or a difference in capture (for example different rolls; mean brightness differs by 1.7 gray levels). Our data has no roll information to separate the two, so we do not claim defect detection beyond "strips containing defects are scored higher".

- **Limitations of the image-level result.** The per-fabric thresholds are computed from patches that do not overlap the ground truth, i.e. they use the masks of the defective images, and operating points are read off on the same defect-free images, so recall and precision are somewhat optimistic. The AUCs do not depend on a threshold.
- **Strict-label results are fragile.** Only 15 images supply strict positives, one image supplies 56% of them, and two of the 15 cross-validation folds contain no strict positives. Type-wise AUCs in Table 8 (e.g. 0.986 for type 036) rest on very few images and should not be over-interpreted.

3 Work To Be Done

- Re-run all experiments (per-fabric models, defect-level evaluation, preprocessing ablation, qualitative examples) on the final sample of all 141 defect-free images, so that every table uses the same data (Table 1).
- Mask blank margins, seams and fabric edges before training, thresholding and scoring, and test the effect on false alarms and on the image-level AUC.
- Image-level: select thresholds without ground-truth masks and on held-out images, and test whether the share score survives matching defective and defect-free strips on brightness within a fabric.
- Add the example figures (cleanest detections, borderline cases, largest misses) to the report.
- Re-run the descriptor selection with relaxed labels and per-fabric models, since the Week 5 ranking used strict labels and a pooled model and was driven by 15 images.
- Improve sensitivity to thin defects: test smaller patches and strides (e.g. 32×32 with stride 16), and features that respond to thin structures, such as maxima or high percentiles of the Gabor response instead of its mean and variance, and row/column profile features for line-shaped defects (hypotheses to test).
- Select thresholds and post-processing on a validation split rather than on the evaluation folds, and report confidence intervals from an image-level bootstrap.
- Follow up on illumination flattening: inspect before/after images for artifacts near defects and borders, try other smoothing scales and flattening followed by CLAHE, and confirm on more images before changing the final preprocessing.
- Identify the 14 four-channel images and check whether the extra channel is constant, to rule out an image-format shortcut.
- Name the 12 defect types from the AFID paper and report results per named type.
- Correct the defect-code and fabric-ID statements and figures from earlier reports in the final report.

4 Literature Review

No change since the mid-semester report: the AFID dataset paper [1], the GLCM [2] and LBP [3] texture-descriptor papers, and a Gabor-filter texture-feature reference [4]. A texture-feature-on-AITEX benchmarking reference remains a candidate addition before end-semester.

5 Challenges Faced

- The 40 defect-free images were sampled from an unsorted file list, so the sample changed between Kaggle sessions and small differences appeared between runs; the build is now deterministic and uses all 141 images.
- Patch scores combined by their maximum do not separate defective from defect-free strips (AUC about 0.5); only the share of flagged patches does (AUC 0.77), and part of that signal could come from capture differences we cannot measure.
- The filename parser used in earlier weeks took the wrong tokens, which invalidated the fabric-ID counts and the "defect code" plots in earlier reports. It was found only by checking the token statistics against the 7 fabric and 12 defect types of AFID.
- The strict 10%-overlap labeling rule ignores thin defects: only 15 of 103 defective images have any strict-positive patch, and one image provides most of them. This makes strict-label metrics unstable and required a relaxed labeling rule.
- Detection of thin defects is weak: at the defect level a 5% false-positive rate gives 43.7 false-alarm patches per image for 37.9% recall, so a purely patch-statistic detector is not yet sufficient.
- Kaggle sessions deleted our cached features twice, so we now back up the cache as a zip file.

References

[1] J. Silvestre-Blanes, T. Albero-Albero, I. Miralles, R. Pérez-Llorens, and J. Moreno, "A public fabric database for defect detection methods and results," *Autex Research Journal*, vol. 19, no. 4, pp. 363–374, 2019.

[2] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural features for image classification," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-3, no. 6, pp. 610–621, 1973.

[3] T. Ojala, M. Pietikäinen, and T. Mäenpää, "Multiresolution gray-scale and rotation invariant texture classification with local binary patterns," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 971–987, 2002.

[4] B. S. Manjunath and W. Y. Ma, "Texture features for browsing and retrieval of image data," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 18, no. 8, pp. 837–842, Aug. 1996.